

Smart Traffic Lights Using Local Artificial Intelligence

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Abstract—Traditional traffic-light systems commonly operate using predetermined timing schedules. Such systems are simple to implement, but fixed timing does not directly respond to changes in traffic conditions. This paper presents a proposed smart traffic-light architecture based on local artificial intelligence (AI), camera-based traffic observation, and edge-side processing. Street cameras provide information about vehicles, while smart pole computers perform local processing and generate traffic control decisions. Selected information may also be transmitted to a central cloud server for storage and wider analysis. The proposed system also considers emergency vehicle detection as an important extension, where nearby traffic signals can be coordinated to provide a clearer route for an emergency vehicle. The paper reviews existing research on AI-based traffic signal control, machine-vision-based adaptive signal control, edge-based traffic control, and emergency vehicle signal preemption. The proposed architecture and control procedure are presented as a research framework; no fabricated experimental performance values are claimed.

Index Terms—Artificial intelligence, smart traffic lights, traffic signal control, machine vision, edge computing, emergency vehicle detection, adaptive traffic control.

I. INTRODUCTION

Traffic congestion is a major challenge in modern urban transportation. Traditional traffic signals generally use predetermined timing plans. Although fixed-time control is simple and inexpensive, it does not directly respond to sudden changes in traffic conditions.

Artificial intelligence (AI) provides an opportunity to make traffic signal control more responsive to observed road conditions. Research has examined the use of deep reinforcement learning for traffic signal control and has identified traffic signal control as an important application area for AI-based methods [1].

Another approach is to use machine vision to obtain information about vehicles from traffic-monitoring cameras. Machine-vision-based research has investigated the use of dynamic traffic-flow information for adaptive traffic-light control [2].

Instead of sending all camera information to a remote cloud server, part of the processing can be performed close to the road using local computing devices. Edge-based traffic-light control has also been investigated, including decentralized reinforcement-learning approaches that operate at the edge of the network [3].

An additional challenge occurs when an emergency vehicle approaches a congested intersection. Emergency vehicle-centered signal control has been studied as a method for reducing emergency vehicle delay while considering the effect on conflicting traffic [4].

Based on these research directions, this paper proposes a conceptual architecture for local-AI-based traffic signals that combines camera-based traffic observation, local processing, adaptive signal control, and emergency vehicle priority.

II. RELATED WORK

A. AI-Based Traffic Signal Control

Rasheed *et al.* presented a review of deep reinforcement learning for traffic signal control. Their work discusses traffic signal control attributes, reinforcement-learning architectures, simulation platforms, complexity, and design considerations [1]. The review demonstrates the growing research interest in applying learning-based methods to traffic signal control.

B. Machine Vision for Adaptive Traffic Signals

Gorodokin *et al.* investigated the optimization of adaptive traffic-light control using machine vision. Their work considers the collection and interpretation of traffic-flow information from monitoring cameras and its use in adaptive signal control [2].

This approach is relevant to the proposed system because cameras can provide information about the current traffic state without requiring manual input.

C. Edge-Based Traffic Control

Zhou *et al.* proposed decentralized reinforcement learning at the edge for traffic-light control in the Internet of Vehicles (IoV). Their approach uses edge servers and decentralized learning for city-scale traffic-light control [3].

The use of edge computing is relevant to local AI because processing closer to the traffic intersection can reduce dependence on a remote processing location.

D. Emergency Vehicle Signal Control

Cao *et al.* proposed an emergency vehicle-centered traffic signal-control approach for intelligent transportation systems. Their work focuses on emergency vehicle performance while

considering the effect of signal preemption on conflicting traffic [4].

This research supports the inclusion of an emergency-vehicle-priority component in the proposed architecture.

III. PROBLEM STATEMENT

Traditional fixed-time traffic signals operate according to predefined timing schedules. Such a system may not respond effectively when traffic conditions change suddenly.

Camera-based adaptive traffic systems can use information from the road to modify signal timing. However, a complete smart traffic system must also consider how quickly the information is processed and how emergency vehicles are handled.

The problem considered in this paper is therefore:

How can local AI and camera-based traffic information be integrated with traffic signals to provide adaptive traffic control while also supporting emergency vehicle priority?

The proposed system is intended to process traffic information locally and coordinate traffic signals when an emergency vehicle is detected.

IV. RESEARCH GAP

Existing research covers several important aspects of intelligent traffic control, including deep reinforcement learning, machine vision, edge computing, and emergency vehicle signal preemption [1]–[4].

The research direction considered in this paper is the integration of these concepts into a single local-AI architecture. In particular, the proposed framework focuses on:

- local processing of camera information;
- adaptive traffic signal decisions;
- reduced dependence on continuous cloud processing;
- emergency vehicle detection;
- coordination of nearby traffic signals; and
- restoration of normal traffic control after an emergency vehicle passes.

The original workshop handout identifies emergency vehicle routing as an important research gap and proposes detecting an emergency vehicle and coordinating nearby signals to create a clearer route. This paper develops that idea as a proposed architecture rather than claiming that the proposed system has already been experimentally validated. [1], [2], [4]

V. PROPOSED SYSTEM

The proposed system uses roadside cameras, local smart-pole computers, traffic signal controllers, and an optional cloud server.

The main processing path is:

Street Cameras → Smart Pole Computers → Decision Engine → Traffic Lights

Selected information may also be transmitted to a cloud server for storage and wider analysis.

A. System Architecture

Fig. 1 shows the proposed architecture.

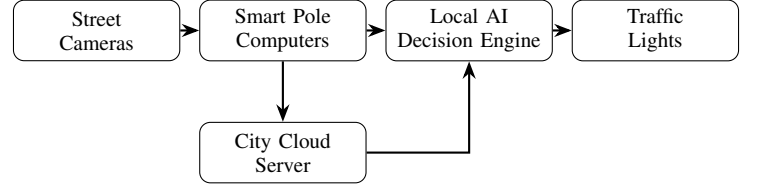


Fig. 1. Proposed local-AI smart traffic-light architecture.

The street cameras capture traffic conditions. The smart pole computers perform local processing of the camera information. The local AI decision engine estimates the current traffic condition and determines an appropriate signal action. The traffic-light controller then applies the decision.

The cloud server is not required to be part of the critical real-time decision path. It may instead be used for storage, historical analysis, system monitoring, or city-level analysis.

VI. INPUT–PROCESS–OUTPUT MODEL

The proposed system can be represented using an Input–Process–Output (IPO) model.

TABLE I
INPUT–PROCESS–OUTPUT MODEL

Input	Process / Output
Camera video	Vehicle detection and traffic estimation
Number of vehicles	Traffic-density estimation
Traffic movement	Adaptive signal decision
Emergency vehicle	Emergency priority decision
Current signal state	Signal coordination and transition
Historical traffic data	Analysis and future optimization

VII. TRAFFIC DENSITY ESTIMATION

The local AI system can estimate traffic density from the number of vehicles detected by the camera.

A normalized traffic-density measure can be defined as

$$D_i(t) = \frac{N_i(t)}{N_i^{\max}} \quad (1)$$

where

- $D_i(t)$ is the normalized traffic density of lane or approach i at time t ;
- $N_i(t)$ is the detected number of vehicles; and
- $N_i^{\max}$ is the selected maximum vehicle count for the observation region.

The value of $D_i(t)$ is normalized between 0 and 1 when $0 \leq N_i(t) \leq N_i^{\max}$.

VIII. ADAPTIVE SIGNAL CONTROL

As a simple proposed control rule, the green duration can be adjusted according to traffic density:

$$G_i(t) = G_{\min} + (G_{\max} - G_{\min})D_i(t) \quad (2)$$

where $G_{\min}$ and $G_{\max}$ represent the minimum and maximum permitted green durations.

For example, a road with higher detected traffic density can receive a longer green interval within the allowed limits.

This equation is a proposed control rule for the framework in this paper. It is not presented as an experimentally validated result.

IX. EMERGENCY VEHICLE PRIORITY

Emergency vehicles such as ambulances may require priority at intersections. Previous research has investigated emergency vehicle-centered traffic signal control and signal preemption [4].

The proposed system adds an emergency vehicle detection stage.

When an emergency vehicle is detected, the system performs the following sequence:

- 1) Detect the emergency vehicle using camera information.
- 2) Determine the direction of travel.
- 3) Identify the upcoming intersection.
- 4) Check the current signal state.
- 5) Generate an emergency priority request.
- 6) Coordinate nearby signals when appropriate.
- 7) Provide a suitable green phase for the emergency route.
- 8) Return the signals to normal adaptive operation.

The emergency state should not simply switch every signal to green. The controller should consider signal safety, conflicting traffic, and the direction of the emergency vehicle.

X. SYSTEM FLOWCHART

The overall operation of the proposed system is shown in Fig. 2.

XI. PROPOSED ALGORITHM

The basic operation of the system can be represented by Algorithm 1.

XII. CONCEPTUAL COMPARISON

The workshop handout provides a simple conceptual comparison between basic timed lights, cloud AI systems, and local smart cameras. Those values are presented in the handout as illustrative comparison values, not as experimentally measured results. Therefore, they are retained here only as an illustrative design comparison and are not claimed as results of this research.

The values in Table II should not be interpreted as experimental measurements. Actual decision latency, internet usage, and implementation cost depend on hardware, software, network conditions, camera resolution, AI model, and deployment environment.

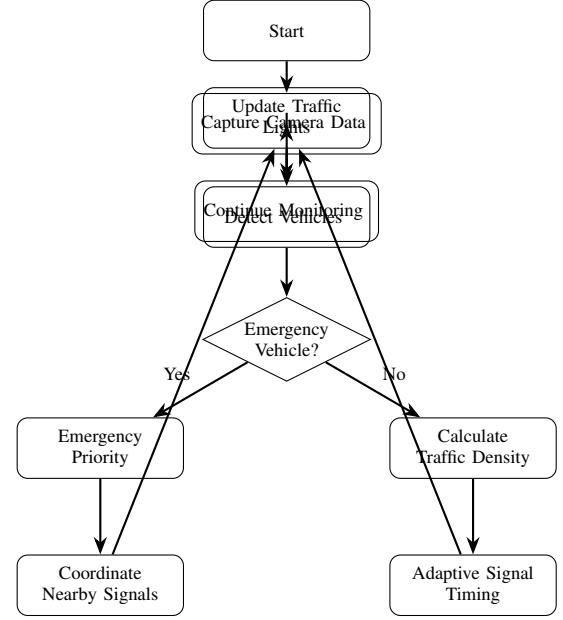


Fig. 2. Flowchart of the proposed local-AI traffic signal system.

Algorithm 1 Local AI Traffic Signal Control

```

1: Initialize traffic signal controller
2: Initialize camera system
3: Initialize local AI processing unit
4: while system is active do
5:   Capture current camera data
6:   Detect vehicles
7:   if emergency vehicle is detected then
8:     Determine emergency vehicle direction
9:     Identify required signal route
10:    Activate emergency priority
11:    Coordinate nearby traffic signals
12:  else
13:    Estimate traffic density
14:    Calculate adaptive green duration
15:    Update normal traffic signal timing
16:  end if
17:  Monitor the updated traffic condition
18: end while

```

TABLE II
ILLUSTRATIVE COMPARISON FROM THE WORKSHOP HANDOUT

System	Decision	Internet	Setup Cost
Basic Timed Lights	30 s	Zero	\$200
Cloud AI	2 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

XIII. ILLUSTRATIVE IMPLEMENTATION

The following C program demonstrates a simplified version of the decision logic. It is only an educational illustration and does not represent a complete traffic-light controller.

Listing 1. Simplified traffic decision logic

```
#include <stdio.h>

int main()
{
    int vehicles;
    int emergency;

    printf("Enter_number_of_vehicles:_");
    scanf("%d", &vehicles);

    printf("Emergency_vehicle_detected?(1=Yes,_0=No)");
    scanf("%d", &emergency);

    if (emergency == 1)
    {
        printf("Emergency_priority_activated.\n");
        printf("Coordinate_traffic_signals.\n");
    }
    else if (vehicles > 20)
    {
        printf("High_traffic_detected.\n");
        printf("Increase_green_time.\n");
    }
    else
    {
        printf("Normal_traffic.\n");
        printf("Use_normal_signal_timing.\n");
    }

    return 0;
}
```

A practical implementation would require camera processing, an AI model, signal-controller interfaces, safety checks, communication protocols, and testing in simulation or a controlled environment.

XIV. EVALUATION PLAN

Because this paper presents a proposed architecture rather than a completed experimental deployment, performance should be evaluated using measurable traffic-control metrics.

TABLE III
SUGGESTED EVALUATION METRICS

Metric	Purpose
Average waiting time	Measure the average delay experienced by vehicles.
Queue length	Measure the number of vehicles waiting at an intersection.
Emergency vehicle delay	Measure the time required for an emergency vehicle to pass through the controlled route.
Non-priority traffic delay	Measure the effect of emergency priority on other road users.
Decision latency	Measure the time between traffic observation and signal decision.
Signal recovery time	Measure the time required to return to normal operation after an emergency event.

Future experiments could compare fixed-time control, adaptive control, and the proposed local-AI approach under identical traffic scenarios.

XV. POTENTIAL ADVANTAGES

The proposed local-AI architecture has several potential advantages:

- **Local processing:** Traffic information can be processed near the road.
- **Adaptive operation:** Signal timing can respond to observed traffic conditions.
- **Emergency priority:** The architecture provides a dedicated mechanism for emergency vehicle detection and signal coordination.
- **Reduced cloud dependence:** Critical traffic decisions can be designed to operate locally.
- **Scalability:** Multiple smart-pole computers can operate at different intersections.
- **Cloud integration:** A central server can still be used for historical data, monitoring, and large-scale analysis.

XVI. LIMITATIONS

The proposed system has several limitations that must be addressed before real-world deployment.

First, camera-based vehicle detection can be affected by lighting, weather, camera position, occlusion, and image quality. Second, an AI model must be evaluated carefully before it is used for safety-critical traffic control.

Third, emergency vehicle detection must be reliable because an incorrect emergency decision could disrupt normal traffic. Fourth, communication between traffic signals must be carefully coordinated to avoid unsafe signal transitions.

Finally, the architecture presented in this paper has not been validated through a real-world deployment. The performance metrics listed in Table III therefore represent an evaluation plan rather than measured results.

XVII. FUTURE WORK

Future work can extend the proposed architecture in several directions.

- 1) Develop and train a camera-based vehicle detection model.
- 2) Develop a reliable emergency vehicle detection method.
- 3) Implement the system in a traffic simulation environment.
- 4) Compare fixed-time, adaptive, and local-AI control.
- 5) Investigate coordination among multiple nearby intersections.
- 6) Measure emergency vehicle delay and normal traffic delay.
- 7) Evaluate the effect of communication latency.
- 8) Investigate privacy-preserving processing of camera data.
- 9) Test the system using a controlled physical prototype.

XVIII. CONCLUSION

This paper presented a proposed architecture for smart traffic lights using local artificial intelligence. The system combines street cameras, local smart-pole computers, an AI decision engine, and traffic signal controllers. The architecture is designed to use traffic information obtained from cameras to support adaptive signal control.

The proposed framework also includes emergency vehicle detection and signal coordination as an important extension. Existing research has already investigated AI-based traffic signal control, machine vision, edge-based traffic control, and emergency vehicle signal preemption [1]–[4].

The main contribution of this paper is therefore the proposed integration of these ideas into a local-AI traffic signal architecture. The system requires further simulation and experimental testing before any claims about actual performance, latency, cost, or emergency response can be made.

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